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Original Research

CeRTS: certainty retrieval token search in large language model clinical information extraction

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ABSTRACT

Objective: Large language models (LLMs) must effectively communicate their uncertainty to be viable in clinical settings. As such, the need for reliable uncertainty estimation grows increasingly urgent with the expanding use of LLMs for information extraction from electronic health records. Previous token-level uncertainty estimators have only used token probabilities within a single output sequence. Here, by leveraging the constraints of JSON output structure, we instead consider all likely sequences and their respective probabilities to obtain a more robust measure of model confidence. We develop Certainty Retrieval Token Search (CeRTS), a new uncertainty estimator for structured information extraction.
Methods: We evaluated CeRTS against a previous gold-standard uncertainty estimator when extracting clinical features from lung cancer discharge summaries across eight open-source LLMs. Calibration (Brier score) and discrimination (AUROC) were used to quantify performance.
Results: CeRTS surpassed the previous gold-standard estimator in discriminatory power across every model and achieved better calibration in most cases. CeRTS had the strongest agreement between model confidence and accuracy with Qwen-2.5.
Conclusion: CeRTS enhances LLM-based information extraction from unstructured clinical text by assigning well-calibrated confidence scores to each extracted item, providing medical researchers with a quantitative measure of reliability at minimal additional cost. Although its performance was generally robust, CeRTS struggled with DeepSeek-R1, which we attribute to the model's Chain-of-Thought reasoning steps. Our evaluation focused on clinical data, but CeRTS can be applied to any domain requiring reliable uncertainty estimation.

1. Introduction

Electronic health records contain a wealth of unstructured information that cannot be directly accessed. The availability of this information significantly enhances medical research and clinical decision-making. Clinical information extraction aims to make this information available in a structured format. [1,2] Structured information is often used in patient phenotyping which has facilitated clinical trial screening, diagnosis categorization, pharmacogenomics, and more [3].

Recent work comparing large language models (LLMs) to rule-based methods has found that they offer competitive performance with significantly fewer pre- and post-processing steps [4]. However, the uncertainty of LLMs is harder to access as this information is encoded across multiple token distributions. Traditional classification approaches to clinical information extraction output class scores that can be interpreted in terms of model uncertainty. The supervised algorithm of PheCAP outputs the predicted probability of patients having a specific phenotype [5]. Similarly, PhenoTagger uses a hybrid approach to identify phenotype concepts by combining an ontology dictionary with BioBERT class prediction [6].

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Without the ability to communicate uncertainty, LLMs will not be leveraged in a clinical context. Uncertainty estimation is necessary to allow medical researchers and physicians to make informed decisions based on extracted patient information. For a confidence estimator to be useful, it must accurately reflect the correctness of the LLM's response. This connection between LLM uncertainty and factual accuracy has been studied across various natural language tasks, including reasoning, question answering, code generation, summarization, and machine translation [7–10]. LLM uncertainty estimation methods fall within three groups: confidence elicitation (CE), sample consistency, and token-level (TL) methods. CE prompts the LLM to include its confidence with its answer on a numeric scale. However, CE was found to be consistently overconfident, greatly reducing its utility in clinical settings [11]. Sample consistency is measured by comparing the agreement between replicate runs of the same prompt. TL methods use the likelihood of generated tokens to measure confidence, where confident answers will have higher token likelihoods.

Recent work highlights a trade-off between the performance of uncertainty estimation and cost. Sample consistency methods had the best performance but were also the most computationally expensive, with Savage et al. using 15 replicate runs, whereas TL methods were the least expensive but performed poorly [11,12]. Consequently, the substantial computational cost of sample consistency methods limits their widespread utility. These studies used individual token probability distributions without considering the entire sequence probability. A separate study improved on TL methods for uncertainty estimation by using logits to estimate uncertainty instead of the normalized probabilities, thereby preserving the overall certainty of the generated sequence at each token [13]. These current TL methods for estimating uncertainty do not utilize the entire token distribution space, highlighting the need for more comprehensive uncertainty estimation techniques. To address this, we introduce Certainty Retrieval Token Search (CeRTS). This uncertainty estimation framework utilizes a structured token search to measure the expected output distribution, followed by a top-2 delta uncertainty estimator, which uses the difference between the two highest output probabilities.

The structured token search has been previously described, where the output distribution was used to calculate the expected value for numerical question-answering tasks to reduce the error of greedy decoding strategies [14]. Because the JSON schema locks in all the fixed tokens (brackets, keys, punctuation), CeRTS only needs to explore the remaining variable fields, letting it systematically sample the model's continuations and approximate the full output distribution. Unlike prior token-level (TL) uncertainty estimators that rely solely on token distributions from individual responses as the variability of the unconstrained output space is too large, CeRTS quantifies model confidence by leveraging the entire output distribution, providing a more comprehensive measure of uncertainty.

Beyond uncertainty estimation, CeRTS captures the expected distribution of alternative LLM responses by exploring multiple token sequences. This process makes uncertainty directly interpretable, as analyzing the labels of these alternative responses can reveal sources of variability.

We aim to further improve the performance of TL methods for structured information extraction while minimizing the computational overhead. We apply CeRTS to clinical information extraction, where ensuring the reliability and transparency of structured data is critical for downstream analysis and decision-making.

Statement of Significance	
Problem to address	Large language models are probabilistic and will not return the same answer when prompted the same way multiple times. Measuring model uncertainty is therefore essential for the application of these models to clinical text by medical researchers.
What is already known	The previous best uncertainty estimator (sample consistency) measures the consistency of responses across 15 runs.
What this paper adds	Our new uncertainty estimator (CeRTS) largely outperforms sample consistency with a much lower computational cost, only requiring an average of 1.9 equivalent runs across all models and features.
Who would benefit from the new knowledge of this paper	Medical researchers performing information extraction with large language models from unstructured clinical notes would greatly benefit. Additionally, CeRTS can be applied to any information extraction task.

2. Methods

2.1. Dataset clinical features

We performed information extraction on discharge summaries for 60 patients diagnosed with lung cancer between 2018 and 2023 from the WashU School of Medicine/BJC Healthcare. The study was approved by the WashU Institutional Review Board (protocol #201905161). These discharge summaries are representative of clinical narratives for information extraction. The clinical features we selected for extraction include the patient's first treatment date, number of discharge medications, mention of lung cancer or carcinoma, immunotherapy treatment, blood pressure value at discharge, and age (Table 1). A Research Electronic Data Capture (REDCap) form was used to create gold-standard labels for each discharge summary by two physicians at the WashU School of Medicine. [15].

2.2. Model setup

Current state-of-the-art API-accessed models do not provide token-level information, limiting our scope to open-source models. We focused on instruction-tuned, lightweight models listed in Supplementary Table 1 [16–23]. These models were 8-bit quantized with float16 precision to reduce GPU resource requirements. This approach provides a baseline for our uncertainty estimator, as we expect larger models to boost accuracy and generate more informative token-probability distributions. We used a sampling temperature of 1.0 and top-p of 1 when measuring sample consistency. When measuring CeRTS confidence, we preserve the original probability distributions to avoid skewing confidence up or down.

2.3. Certainty retrieval token search (CeRTS)

CeRTS is a two-step approach for estimating LLM uncertainty (Fig. 1). It begins with a structured token search, which identifies high-probability output sequences, followed by top-2 delta, an uncertainty estimator using the probability difference between the top two output sequences.

Table 1
Annotator Cohen's Kappa and Intraclass Correlation Coefficient (ICC) scores of extracted clinical features.

Feature	Metric	Value
Age	ICC	0.993
Number of Discharge Medications	ICC	0.964
First Treatment Date	Cohen's Kappa	0.617
Mention of Lung Cancer/Carcinoma (Yes/No)	Cohen's Kappa	0.551
Receiving Immunotherapy (Yes/No)	Cohen's Kappa	0.732
Blood Pressure	Cohen's Kappa	0.931

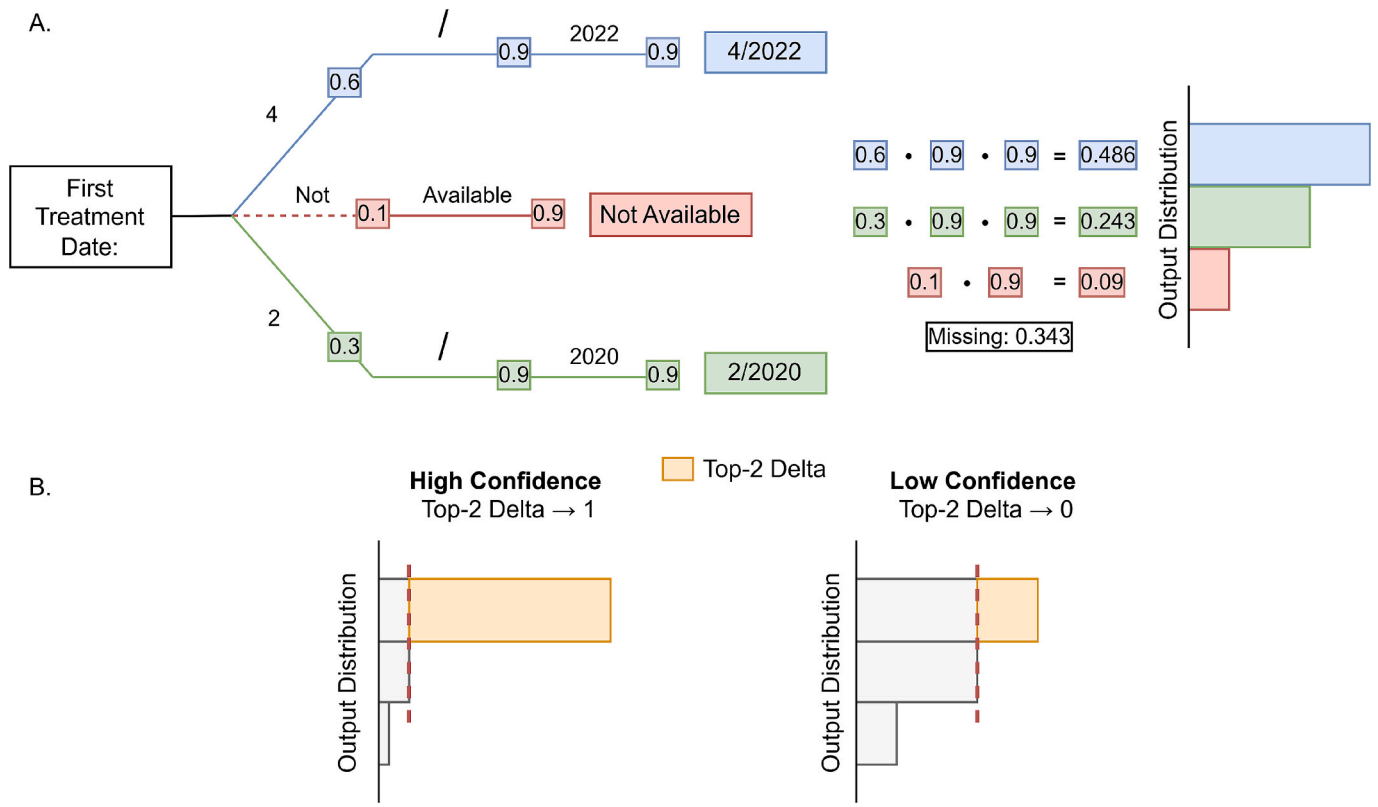


Fig. 1. Illustration of CeRTS when extracting the first treatment date. (A) Structured Token Search: The output distribution is measured by the product of conditional probabilities for each output sequence. (B) Top-2 Delta: LLM confidence is measured using the probability difference between the two most likely output sequences (dashed red line). An LLM is more confident when the output distribution is dominated by a single response with the top-2 delta approaching 1 (orange boxed region). (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

2.4. Structured token search

The search is performed by exploring a priority queue that orders subsequent tokens by their logits. Next, tokens that fall below the specified probability threshold are not added to the priority queue. Here, we selected a minimum sequence probability threshold of 0.05, as this offered a good trade-off between output distribution completeness and runtime during our preliminary investigation. The maximum number of inference steps was set to 100 to limit the search space. Fig. 1 outlines the complete workflow for our uncertainty estimator. After estimating the full output distribution, a "missing" term represents the probability of all sequences that fall below the set probability threshold.

2.5. Uncertainty estimator selection

After measuring the expected output distribution with the structured token search, we applied multiple certainty metrics and compared their performance to select a final uncertainty estimator for CeRTS. The certainty metrics we explored include top-2 delta, top-1 probability, minimum token probability, and normalized entropy. The top-2 delta is measured as the difference between the probabilities of the top two most likely answers (Fig. 1b). If a single highly likely answer dominates the distribution, the top-2 delta approaches 1, indicating high confidence. Conversely, if multiple top responses have similar probabilities, the top-2 delta approaches 0, indicating greater uncertainty. The top-1 probability directly uses the probability of the most likely answer. Similarly, the minimum token probability uses the probability of the least likely token in the primary answer. If only one sequence is generated, then the top-2 delta uses only the top-1 probability. Lastly, we define the normalized entropy in Eq. (3) as one minus the ratio between the entropy (H) of the expected output distribution (p) to the entropy of a uniform distribution (u) with the same number of elements N.

$$H(p) = -\sum_{i=1}^{N} p_i log p_i \tag{1}$$

where $p = (p_1, \cdots, p_N)$ is a discrete probability distribution, $p_i \geq 0$ and $\sum_i p_i = 1$. The uniform distribution over N outcomes is.

$$u_i = \frac{1}{N}, i = 1, \cdots, N \tag{2}$$

$$NormalizedEntropy(p, N) = 1 - \frac{H(p)}{H(u_N)} \tag{3}$$

We used $N = 15$ as this captured the maximum number of responses across models. To account for the missing probability density from each structured token search, the remaining elements were filled equally with any remaining density. When a model failed to return a JSON output, the response was labeled as 'missing', and each certainty metric was set to 0.

2.6. Sample consistency uncertainty estimator (SC)

We adapted the Sample-Consistency (SC) uncertainty estimator from the best-calibrated method reported previously (Fig. 2). [11] For each patient feature, the same extraction prompt is provided to the LLM 15 times. The model's confidence is then quantified as the proportion of outputs that match the most frequent response, where high agreement implies high confidence. To identify the modal answer, the same LLM re-evaluates its own 15 outputs in an "annotator" role and returns a JSON object specifying the most frequent response and its count. A temperature of 1 is used during sampling to ensure adequate variability in responses. The prompt provided to the annotator includes a worked example (blood-pressure extraction) to illustrate the expected JSON

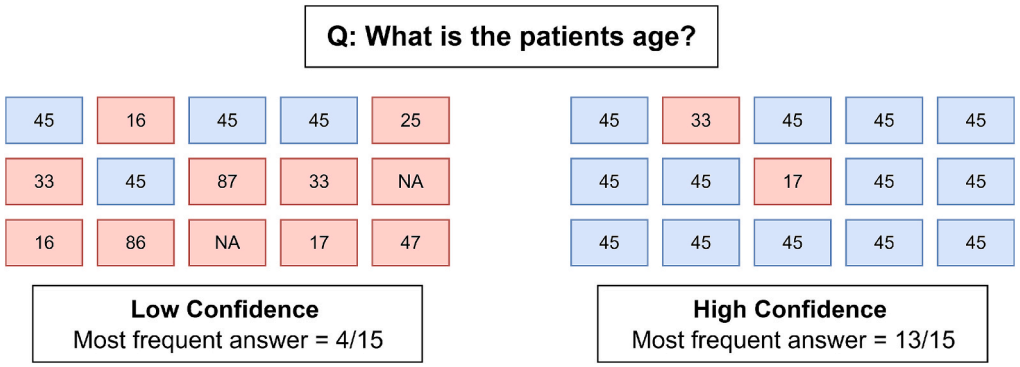


Fig. 2. Example of measuring sample consistency (SC) confidence for extracting patient age. Multiple answers are recorded from the same prompt and compared by an LLM annotator to determine the count of the most frequent answer.

format.

2.7. Evaluation of uncertainty estimators

CeRTS and SC are evaluated in terms of discrimination and calibration (Fig. S1). Discrimination measures how well the uncertainty estimator separates incorrect and correct outputs and is measured by the area under the receiver operating characteristic (AUROC). Calibration measures how well changes in uncertainty are reflected in output accuracy and is measured by the Brier score, defined as:

$$BrierScore = \frac{1}{N}\sum_{i=1}^{N}(confidence(\widehat{y}_i) - 1(\widehat{y}_i = y_i))^2 \quad (4)$$

The Brier score measures the average agreement between the confidence estimator and accuracy across all samples. [24] Where the confidence of the answer $\widehat{y}_i$ is compared against the true label y_i. Brier scores closer to 0 indicate better agreement between CeRTS confidence and model accuracy. The key difference between these metrics is that discrimination only requires that some arbitrary threshold separates incorrect and correct outputs, while calibration requires that uncertainty estimates are directly interpretable in terms of accuracy.

2.8. Prompt design

An example of how the LLM is prompted to extract each clinical feature into JSON format is shown in Fig. S2. The same prompt template was used for each feature and model. The prompt template requires the name of the feature, a JSON output example, and a JSON output example if the information is missing. This prompt template ensured reliable extraction performance by minimizing variability in the output space. The JSON output example for missing information varied depending on the task. For mention of lung cancer, we specify 'NO', and for number of discharge medications we specify '0′ if the information is missing. For the remaining features, including blood pressure, age, and first treatment date, we specify 'Not Available'. CeRTS facilitated prompt design by analyzing the output distribution, allowing for the refinement of prompts that introduced significant variability in model responses.

2.9. Experimental overview

We first assessed each model's accuracy and instruction-following ability for every clinical feature, because models that frequently omit an answer or give an incorrect one can still show strong agreement with uncertainty metrics. These baseline results provide a foundation for interpreting the performance of our uncertainty-estimator. The highest probability output from each structured token search was selected as the model's response. Accuracy was evaluated by comparing model-generated responses to gold standard labels for each discharge summary. In most cases, correctness was determined using exact string matching. The only exception was the 'first treatment date' field, where additional normalization was applied to account for variations in date formatting. Specifically, dates were parsed and standardized to ensure equivalence regardless of whether they were expressed in MM/DD/YY or MM/DD/YYYY format, with two-digit years assumed to belong to the 2000 s. Invalid or unparsable dates were treated as mismatches. The instruction-following ability of each model was evaluated using the missing probability density from structured token search. The missing probability density represents sequences where an end-of-sequence token is not generated before the sequence probability falls below the minimum threshold (0.05) or reaches the maximum number of tokens (100).

The performance of each uncertainty estimator was measured in terms of calibration and discrimination. We first compared multiple confidence metrics to select the strongest performer for CeRTS. These metrics included top-2 delta, top-1 probability, minimum token probability, and normalized entropy. SC was then evaluated against CeRTS to assess its effectiveness for each clinical feature and model. Calibration plots with 10 bins of equal size were used to visualize how well uncertainty values aligned with the observed accuracy. To assess overconfidence in CeRTS and SC, an overconfidence score was calculated as the difference between average confidence and the overall accuracy for each feature and model. Positive values indicate overconfidence, where the model's confidence exceeds its actual accuracy, while negative values indicate underconfidence.

To evaluate the downstream application of the CeRTS uncertainty estimator with a specific model and clinical feature, we plotted the false-positive rate and false-negative rate while varying the CeRTS confidence threshold. These results demonstrate how a confidence threshold can be selected based on the task's requirements.

To quantify the extra computation required by CeRTS over standard sampling, we define the number of equivalent runs: the number of ordinary, single-pass generations needed to produce the same total token count as one structured token search. Formally, we divide the total tokens created during the search by the mean tokens per response for the relevant model–feature pair. We use the mean because individual responses vary in length, relying on any single example would misrepresent the typical token count. If a structured token search was terminated before generating an end-of-sequence token, a sample response was used to approximate the average token length.

3. Results

3.1. Clinical information extraction performance

Individual model performance for each clinical feature was evaluated by comparing predicted outputs to the gold standard labels from

each discharge summary. The highest probability output from each structured token search was selected as the model's response. The first treatment date and the number of discharge medications were the most challenging features to extract across all models (Fig. 3a). Qwen-2.5 achieved the highest accuracy with the first treatment date (0.67), while Mistral performed best with the number of discharge medications, with an accuracy of 0.30. In contrast, extracting blood pressure at discharge and patient age were generally the easiest tasks, with most models achieving accuracies above 0.90. However, DeepSeek-R1 struggled with extracting blood pressure, achieving an accuracy of just 0.32. Qwen-2.5 had the most consistent performance across all features, consistently ranking among the top-performing models.

The consistency of each LLM in returning the expected JSON format is referred to as its "instruction-following ability". The instruction-following ability of each model was evaluated by the missing probability density from each structured token search (Fig. 3b). A large amount of missing probability density indicates that the model frequently does not return the correct JSON format. DeepSeek-R1 would rarely reach an end-of-sequence token before all sequences fell below the minimum probability threshold. DeepSeek-R1 was only able to provide a response when it answered without generating any thought tokens. Mistral, Phi-4, and Gemma-2 had high missing probability densities, typically ranging from 0.3 to 0.5. Qwen-2.5 and Hermes-3 demonstrated the strongest instruction-following ability, with most values being near zero.

3.2. Uncertainty metric selection for CeRTS

We evaluated the performance of four uncertainty metrics across eight language models using output distributions obtained from the structured token search (Supplementary Table 2). Performance was first assessed by aggregating predictions across clinical features and

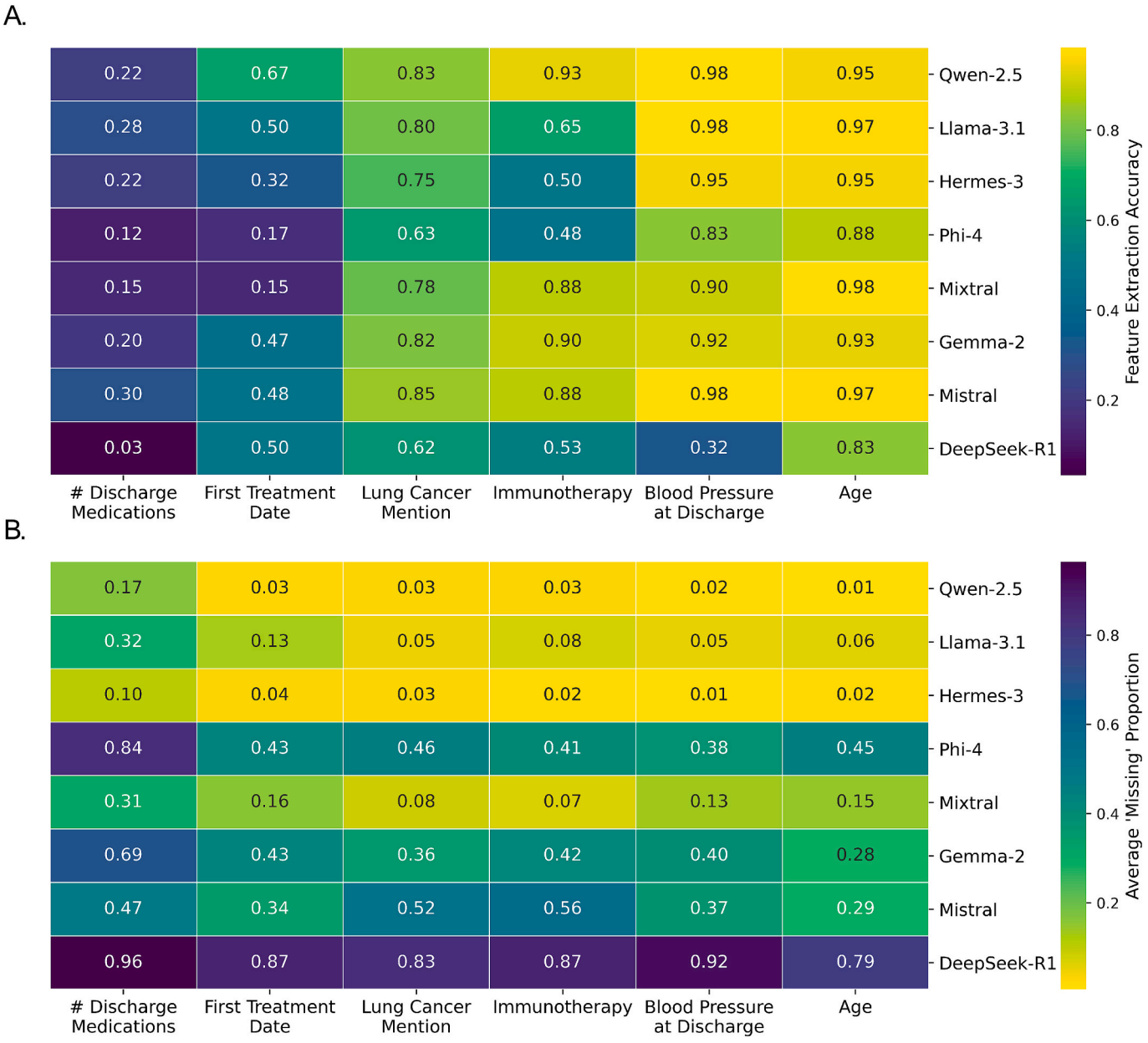


Fig. 3. Clinical information extraction performance. (A) Heatmap of model accuracy by clinical feature. (B) Heatmap of the average proportion of missing probability density in the output distribution from each Structured Token Search. A low proportion of missing probability density (yellow) indicates more consistent instruction following. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

evaluating both calibration and discrimination for each model. Top-2 delta achieved the best discrimination in six out of eight models and the best calibration in two models. Top-1 and normalized entropy showed comparable calibration performance, with each ranking highest in three and two models, respectively. However, both models achieved the best discrimination in only one case. Minimum token probability consistently underperformed, with particularly poor calibration and discrimination, with Gemma-2 having a Brier score of 0.49 and AUROC of 0.54.

We then assessed aggregate performance by combining all predictions and corresponding confidence scores across the eight models. Top-2 delta achieved the best performance with a Brier score of 0.183 and AUROC of 0.82. Top-1 was the second-best performer with a Brier score of 0.184 and AUROC of 0.79. Given its consistently strong performance across individual models and in aggregate, top-2 delta was selected as the uncertainty metric for CeRTS.

3.3. Uncertainty estimator evaluation

Uncertainty estimators were evaluated in terms of discrimination (AUROC) and calibration (Brier score). The Brier score measures the direct alignment between confidence and accuracy, ranging from 0 to 1, with a value near 0 indicating the best alignment. To measure the overall performance of each uncertainty estimator, we aggregated all responses across clinical features for each model. Responses from different models were evaluated separately and not combined. Performance metrics computed over the combined responses for each model are summarized in Table 2. CeRTS outperformed SC in terms of discrimination across all models, with there being a relatively large gap of 0.16 in AUROC for Hermes-3. CeRTS outperformed SC in terms of calibration in 5 out of 8 models, with moderate performance improvements. SC had significantly better calibration with DeepSeek-R1.

CeRTS performed the best with Qwen-2.5, achieving an aggregate Brier score of 0.10 and AUROC of 0.89. In contrast, DeepSeek-R1 showed the weakest calibration, with a higher Brier score of 0.30, indicating poor alignment between predicted confidence and actual correctness. Despite this, DeepSeek-R1 achieved the highest AUROC of 0.92, suggesting that its confidence scores still effectively distinguished between correct and incorrect responses when a threshold was applied. The model's low CeRTS confidence values contributed to its poor calibration, but these scores remained informative when used for classification. The majority of the best feature-specific Brier scores were achieved with Qwen-2.5 (Fig. S3a), except for the number of discharge medications, where other models performed marginally better. Qwen-2.5, Llama-3.1, and Hermes-3 had near-zero Brier scores for age and blood pressure at discharge, indicating extremely high agreement between CeRTS confidence and model accuracy. Feature-specific AUROC values showed much less consistency across top models (Fig. S3b). CeRTS struggled to separate incorrect and correct answers with Qwen-2.5 and Llama-3.1 for blood pressure, immunotherapy, and mention of lung cancer.

Table 2
Aggregate performance of the CeRTS and SC uncertainty estimators across all answers. Uncertainty values and corresponding correctness for all model outputs were used to measure comprehensive performance. Brier scores near 0 indicate high agreement between model uncertainty and correctness.

	Calibration (Brier Score)		Discrimination (AUROC)	
Model	CeRTS	SC	CeRTS	SC
Qwen2.5-7B-Instruct-1 M	0.10	0.10	0.89	0.86
Mixtral-8x7B-Instruct-v0.1	0.14	0.19	0.83	0.77
Gemma-2-9b-it	0.15	0.14	0.86	0.83
Llama-3.1-8B-Instruct	0.15	0.16	0.84	0.83
Phi-4-mini-instruct	0.24	0.26	0.77	0.72
Mistral-Small-24B-Instruct-2501	0.20	0.12	0.87	0.86
Hermes-3-Llama-3.1-8B	0.19	0.24	0.87	0.71
DeepSeek-R1-Distill-Llama-8B	0.30	0.15	0.92	0.84

3.4. CeRTS overconfidence

Overconfidence in an uncertainty estimator greatly reduces its clinical utility. Calibration plots for each model when extracting the number of discharge medications provide a more detailed interpretation of the calibration and discrimination results (Fig. 4a). These plots visualize the agreement between model confidence and accuracy. Importantly, CeRTS correctly identifies samples with very low and high accuracy, as seen by the values near (0,0) and (1,1). Values below the perfect calibration line indicate CeRTS overconfidence. CeRTS had less overconfidence than SC (Fig. S5b). The overconfidence of CeRTS varies significantly across models (Fig. 4b). Mixtral and Phi-4 were underconfident in their predictions for blood pressure and age, but were significantly overconfident when predicting the first treatment date, with overconfidence scores of 0.39 and 0.26, respectively. The CeRTS confidence of Qwen-2.5 and Llama-3.1 closely matched their accuracy, with slight overconfidence for most features. The number of discharge medications featured strong alignment between CeRTS confidence and accuracy for all models. CeRTS and SC both have the greatest overconfidence when extracting the first treatment date with Mixtral, Hermes-3, and Phi-4.

We inspected the notes that had the greatest CeRTS overconfidence and found that Hermes-3 had often selected 'YES' for receiving immunotherapy when the correct answer was 'NO'. We could not identify a cause for these misclassifications within the notes.

3.5. Cost analysis of the structured token search

The SC method requires 15 replicate runs for each run, resulting in a significant computational cost. To measure the additional cost of CeRTS relative to standard sampling, we compare the number of inference steps required for each structured token search to the number of tokens in a singular response (Fig. 5). The average number of equivalent runs was measured as the ratio between the number of tokens generated during the structured token search and the average number of tokens in each response. As expected, the computational cost increases as model confidence decreases since a greater search space is required. We find that even for features with high degrees of uncertainty, the number of equivalent runs does not increase significantly and plateaus around 2 for most models. Qwen-2.5 and Hermes-3 had the lowest average cost for each structured token search, with most features remaining near 1. The average number of equivalent runs across all models and features is 1.9.

3.6. Balancing FPR and FNR in CeRTS-driven extraction

To achieve the desired false-positive (FPR) and false-negative (FNR) rates in a clinical setting, a confidence threshold can be first calibrated on a labeled subset and then applied to the remaining unlabeled cases. To be clinically useful, an uncertainty estimator needs to reach a relatively low proportion of incorrect answers that it labels trustworthy (FPR). To evaluate the downstream application of CeRTS, we measured the FPR and FNR while varying the CeRTS confidence threshold with Qwen-2.5 (Fig. 6). The threshold required to reach a FPR $\leq$ 0.10 was 0.19 for the number of discharge medications, at which point the FNR was 0.46. When extracting the first treatment date, the same FPR was achieved only at a confidence of 0.98, with a FNR of 0.49. Improving the first treatment date FPR any further would cause most correct answers to be discarded.

4. Discussion

We propose CeRTS for estimating uncertainty in structured information extraction. We evaluated this new uncertainty estimator across

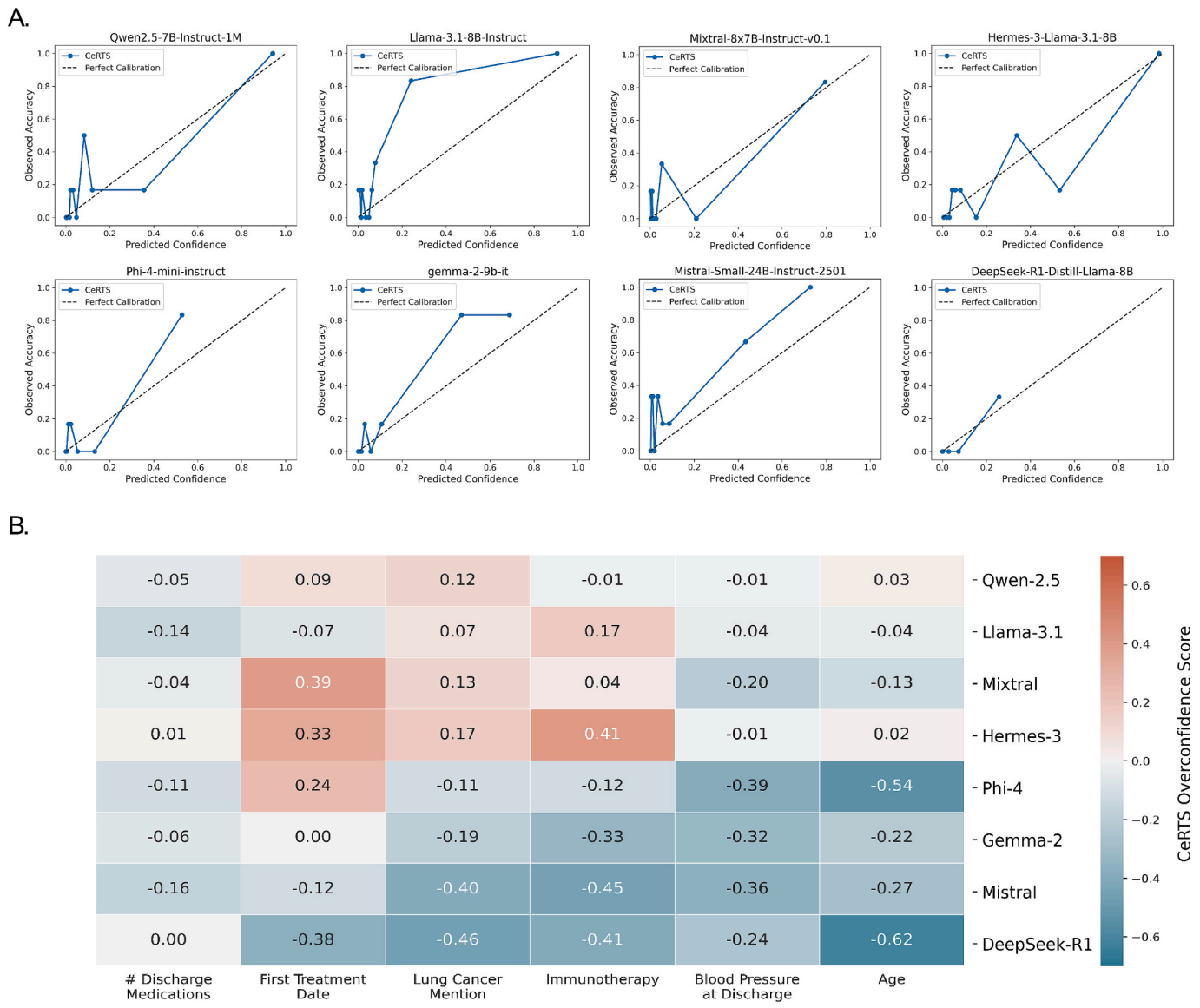


Fig. 4. Comparison of CeRTS confidence to model accuracy (A) Calibration plots of CeRTS predicted confidence against the observed accuracy for extracting the number of discharge medications. Calibration plots were created with 10 confidence bins of equal size. (B) Heatmap of overconfidence scores, measured by the difference between CeRTS confidence and the observed accuracy. Larger values (red) indicate overconfidence and smaller values (blue) indicate underconfidence. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

eight language models, assessing its performance in terms of discrimination and calibration. We found that CeRTS consistently outperformed the previously described gold-standard SC method [11] with greatly reduced computational cost.

The agreement between the uncertainty estimators and model accuracy was measured by Brier score and AUROC. Since SC tallies how often the modal answer recurs across replicates, it is very similar to top-1. This similarity is reflected in their performance as both SC and top-1 provide modest calibration but lag behind top-2 delta in discrimination. During our evaluation of alternative uncertainty metrics for CeRTS, normalized entropy performed worse than top-2 delta and top-1 despite leveraging the full output distribution. This suggests that the probabilities of the two most likely sequences already capture most of the model's uncertainty, though a systematic exploration of additional full-distribution transformations may still reveal further improvements.

CeRTS and SC displayed strong calibration, with most Brier scores being below 0.2 (Figs. S3a and S4a). SC had poor discrimination for a majority of feature and model combinations. Previous work found that measuring the agreement of the 15 responses using cosine similarity yielded the best discrimination, accompanied by a significant drop in calibration. [11].

Extracting the first treatment date had the worst agreement between confidence and accuracy for both uncertainty estimators (Figs. S3 and S4). This may reflect a data quality issue, as the inter-annotator agreement was relatively low with a Cohen's Kappa of 0.617 for this feature.

Both uncertainty estimators struggled when there was an extreme class imbalance of incorrect and correct responses, as seen by the respective accuracy values. To account for this class imbalance, we measured the aggregate performance of CeRTS and SC across all responses for each model (Table 2). CeRTS performed the best with Qwen-2.5, achieving an aggregate Brier score of 0.1 and AUROC of 0.89.

CeRTS had less overconfidence than SC, indicating that it produces more trustworthy confidence scores. This is an essential requirement for clinical decision support, where overconfident errors can mislead clinicians and ultimately compromise patient safety (Figs. 4 and S5b). Hermes-3 had the greatest CeRTS overconfidence when identifying if the patient was treated with immunotherapy. Improved prompting that defines different forms of immunotherapy is likely needed. Few-shot

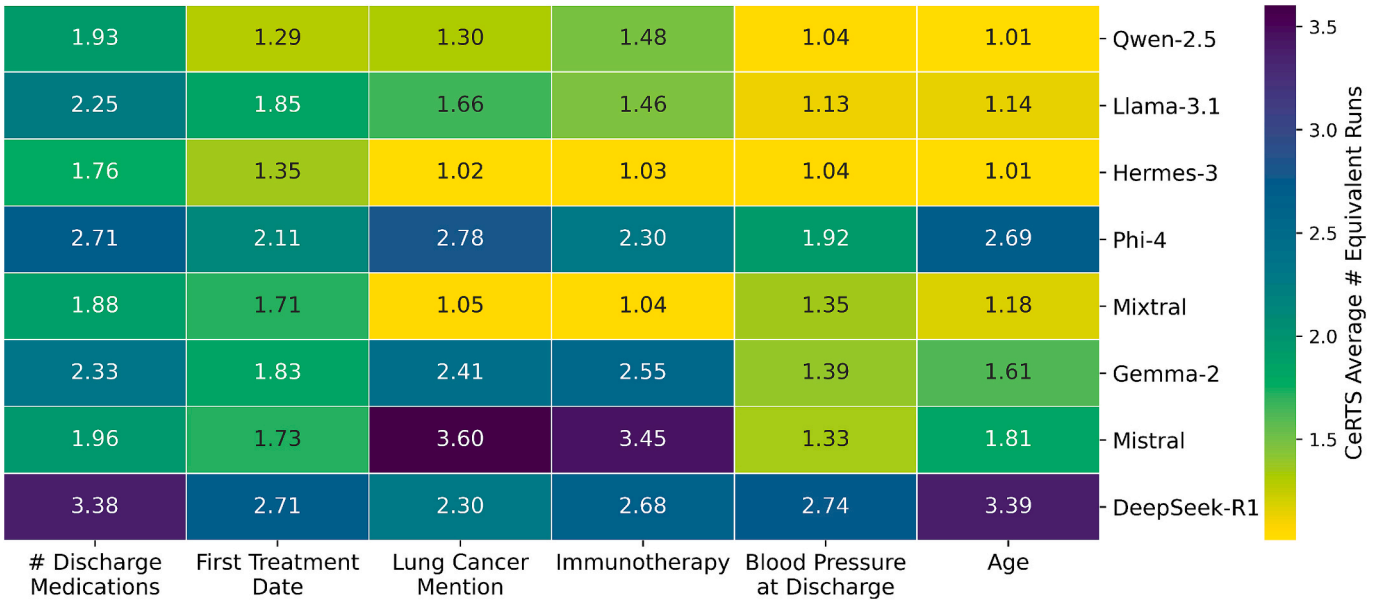


Fig. 5. Heatmap of the relative cost of each Structured Token Search compared to generating a single response. The relative cost, expressed as # Equivalent Runs, represents the number of inference steps required by each Structured Token Search, normalized by the steps needed to produce a single response.

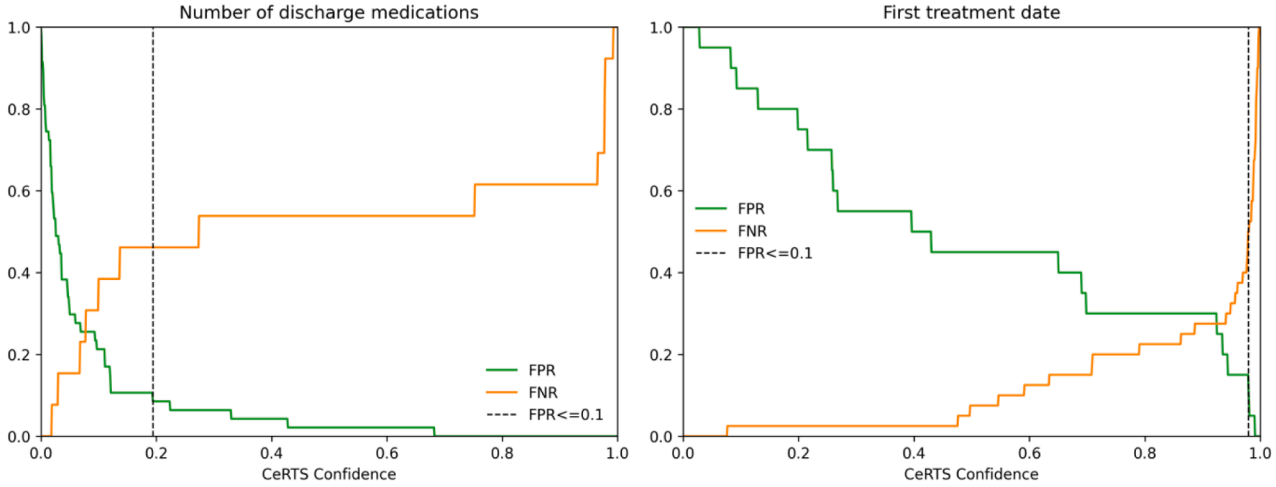


Fig. 6. False positive rate (FPR) and false negative rate (FNR) plotted against CeRTS confidence threshold with Qwen-2.5. Answers above the confidence threshold are considered trustworthy. The FPR is the proportion of incorrect answers that were considered trustworthy.

prompting may also provide a calibration boost to CeRTS and should be explored in future work.

While further prompt refinement may reduce the proportion of missing probability density for some models, CeRTS performance remains robust overall. Notably, models with higher average missing probability density, such as Gemma-2, still rank among the top performers (Table 2).

We quantified the cost of CeRTS by counting the inference steps in each structured token search (Fig. 5). For most models and features, the number of equivalent runs remained below 2, indicating that CeRTS maintains a low computational cost. Notably, this is much lower than SC, which requires 15 samples in addition to LLM annotation prompting. The ability of CeRTS to measure the full output distribution comes from its constrained search space. Alternative tokens are not explored when generating the JSON structure, including the opening and closing brackets, keys, and any fixed string elements. However, repeated runs of SC will need to regenerate these tokens. Additionally, the computational cost of CeRTS can be further reduced by performing a beam search for the top-2 sequences instead of traversing additional sequences, as here we measure the entire output sequence above the probability threshold of 0.05.

CeRTS improves the reliability of extracted information for use in medical research. After using a small set of labeled examples to choose an appropriate confidence threshold, researchers can apply that cut-off to the remaining unlabeled data to filter or flag uncertain predictions according to their study's needs as seen in Fig. 6, where a low FPR or FNR can be prioritized depending on the medical researcher's needs. With this, CeRTS strengthens the trustworthiness of downstream analyses by quantifying the reliability of clinical data extracted from unstructured text.

CeRTS is not limited to clinical information and can be applied to any structured information extraction task. Additionally, the structured token search provides a cost-effective method for approximating output distributions of closed-source, API-accessed models, especially as token-level distribution information becomes increasingly available. In the context of clinical notes, high uncertainty in an extracted feature, such

as a date, may indicate that multiple dates are mentioned. The complete set of dates present in the output distribution can then be examined for validation, ensuring greater transparency and reliability in the information extraction process. Interpreting the output distribution is also especially important when multiple correct answers exist, as the probability of each synonymous response may be nearly equal, resulting in very low confidence. A human-in-the-loop or LLM can be used to identify these cases, and the probabilities of synonymous responses can be combined before computing the top-2 delta.

Further work is needed to validate the performance of CeRTS in other information extraction tasks. We have only explored the use of the structured token search where one entity is extracted at a time. To address multi-entity extraction, we plan to only explore the token space in variable JSON regions. Entity dependencies can significantly impact the distributions of later fields and hide model uncertainty. These dependencies can be detected by measuring the mutual information between JSON entities. If there are no significant entity dependencies, greedy sampling can be used as the JSON is generated to fix previous fields to their top answers. To handle entity dependencies, we can compute the marginal distribution with the sum of conditional probabilities for a given entity value. The two highest probabilities in the resulting marginal distribution can then be used to measure CeRTS confidence.

5. Conclusions and limitations

CeRTS enhances LLM-based information extraction from unstructured clinical text by assigning well-calibrated confidence scores to each extracted item, providing medical researchers with a quantitative measure of reliability at minimal additional cost. Although its performance was generally robust, CeRTS struggled with DeepSeek-R1, which we attribute to the model's Chain-of-Thought reasoning steps. Additionally, the confidence estimator was only applied to one type of clinical text, which may not generalize to all types of clinical text or other information extraction tasks.

6. Code availability

Our implementation of CeRTS is available at https://github.com/LarsSchimmelpfennig/CertaintyRetrievalTokenSearch.

Declaration of Generative AI and AI-assisted technologies in the writing process

Writing was proofread by Grammarly followed by review and editing by the authors. The authors take full responsibility for the content of this document.

CRediT authorship contribution statement

Lars E. Schimmelpfennig: Writing – review & editing, Writing – original draft, Methodology, Conceptualization. **Kriti Bhattarai:** Data curation. **Inez Y. Oh:** Writing – review & editing. **Jake Lever:** Writing – review & editing. **Obi L. Griffith:** Writing – review & editing. **Malachi Griffith:** Writing – review & editing. **Albert M. Lai:** Writing – review & editing. **Zachary B. Abrams:** Writing – review & editing, Supervision.

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Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Appendix A. Supplementary data

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